Capital discipline

Combined factor-model portfolio test

Practitioner factor-model runner

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Abstract

The run combines 2 normalized factor scores (Investment, ExternalFinance) into one stock-level model score before constructing one portfolio on 300 sampled symbols from gold version=20260822T141457Z. The combined net model earns -1.09% annually on an arithmetic basis and -2.61% compounded (the figure the growth chart shows), with 17.64% volatility, Sharpe -0.06, and HAC t=-0.24. The report is generated from the exact configuration and measured outputs of this run; it is an exploratory model diagnostic, not evidence from an untouched confirmatory sample.

1 Research objective

This run asks whether the configured factor model produces a useful portfolio return and whether that model return is explained by the supplied benchmark or standard factor model. Component signals are observed at month-end and evaluated on next-calendar-month simple returns.

Construction rule. Each signal is normalized within month, component signals are weighted equally inside each named factor unless their definitions say otherwise, and normalized factor scores are combined at the stock level using Investment 50%, External Finance 50%. The runner then constructs one model portfolio, applies beta neutrality once if requested, and charges costs once. It does not average independently constructed factor-portfolio returns.

2 Configuration and sample

Factor	Composite signals	Signal transform	Composite transform	Winsor	Min inputs
Investment	asset_growth_yoy	rank	rank	1%-99%	1
External Finance	equity_issuance_yoy, debt_issuance_yoy	rank	rank	1%-99%	2

Model construction	Book	Beta neutral	Bucket frac.	Cost bps	Min factors	Benchmark
signal weighted	long-short	no	0.33	0.0	2	SPY

Portfolio	Mean stocks	Minimum stocks
Combined model	36.1	20
Investment	38.8	21
External Finance	36.1	20

The screened sample contains 300 symbols and 8,850 stock-months from 2005-09-01 through 2026-08-01. Universe settings are: minimum as-traded price 1.0, minimum 21-day dollar volume None, minimum market capitalization None, top-cap fraction 0.375, and financial exclusion True. Benchmark configuration is symbol SPY or external file none. Standard-factor controls use file standard_recreated_factors.csv. The runner reports the supplied files verbatim and does not infer their provenance.

3 Portfolio returns

What each book earned, raw. A standalone sleeve is the same construction as the combined model run on a single factor, and the existing model is the incumbent all of them have to beat, so every row is comparable on one shared window. Return and volatility use monthly arithmetic moments, Sharpe is their ratio, *t* is HAC. Statistics are in the table throughout this report; figure legends name their lines and nothing else.

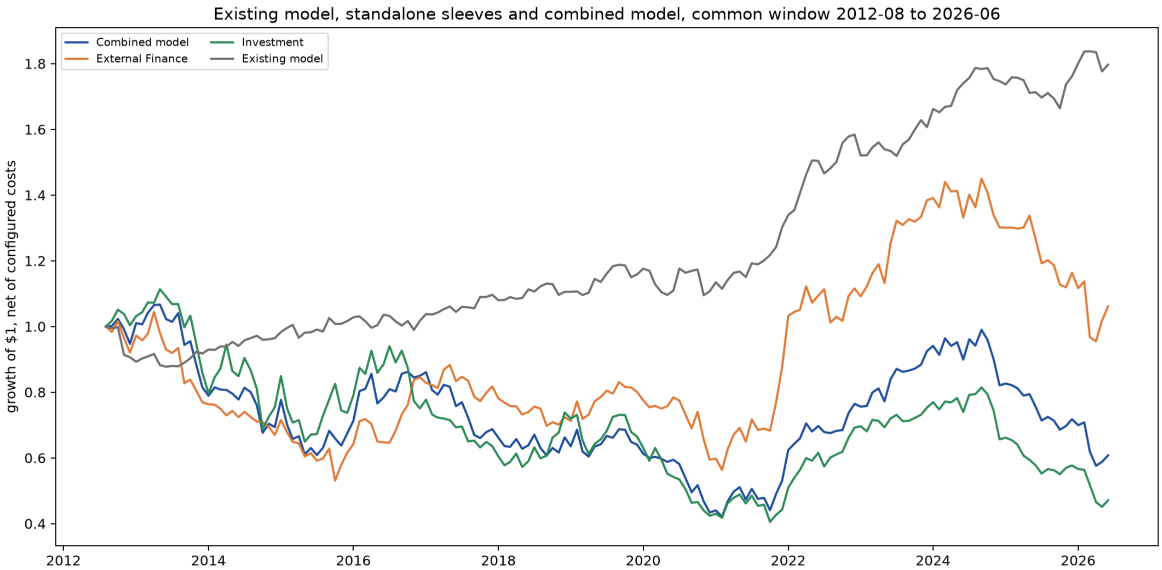

Figure 1: Growth of one dollar in each book, net of configured costs, on the common window. The path begins at one before the first return month.

Portfolio	Months	Ann. ret.	CAGR	Ann. vol.	Sharpe	t
Combined model	197	-1.09%	-2.61%	17.64%	-0.06	-0.24
Investment	203	-2.38%	-3.97%	18.21%	-0.13	-0.51
External Finance	197	2.35%	0.83%	17.51%	0.13	0.52
Existing model	201	4.54%	4.42%	6.36%	0.71	2.89

4 Benchmark fit and benchmark-orthogonalized returns

Each raw net return from Section 3 is regressed only on the SPY market benchmark. The column headed “R2: SPY market benchmark” is the share of that portfolio’s monthly net-return variance explained by the SPY market benchmark; it is not the R2 of the standard-factor model. Alpha is the annualized mean return left after removing the benchmark slope, *t* is its HAC statistic, and F tests the benchmark slopes jointly. The combined model’s estimated MKT loading is -0.20. Because that loading is negative, removing the fitted benchmark component can raise the orthogonalized return above the raw return when the benchmark rises. This is an alpha-preserving diagnostic return, not a costless improvement to the raw portfolio.

Portfolio	N	Loading (MKT)	t (loading)	R2 (SPY market benchmark)	F
Combined model	197	-0.20	-2.13	0.025	4.55
External Finance	197	-0.15	-1.54	0.015	2.38
Investment	203	-0.25	-2.60	0.039	6.74
Existing model	230	-0.11	-2.36	0.060	5.58

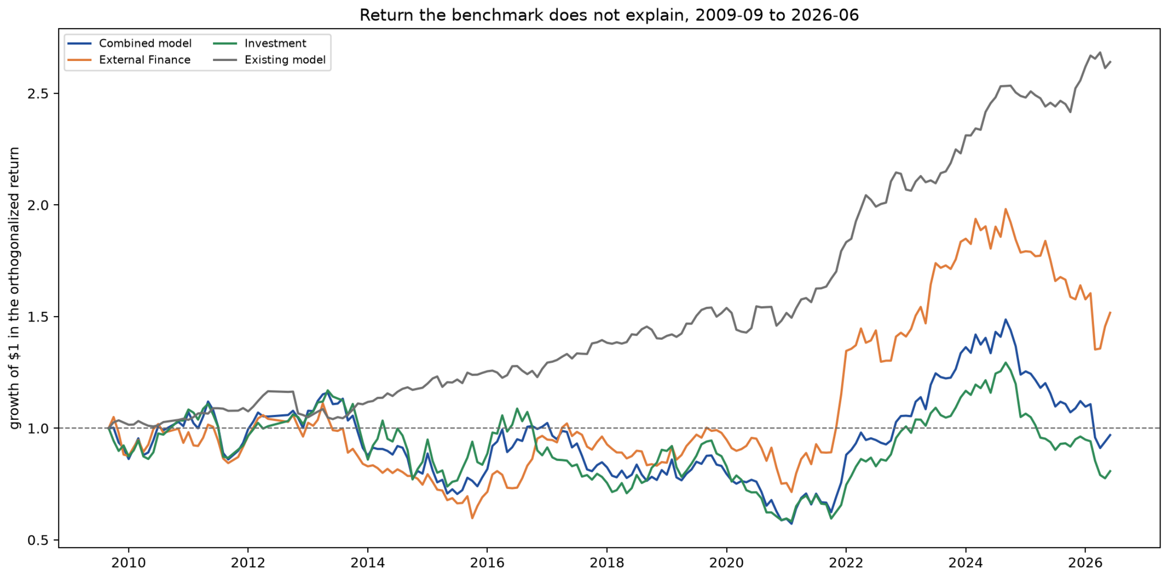

Figure 2: Returns orthogonalized to the SPY market benchmark: growth of one dollar in each book once the SPY market benchmark component is removed. A flat path earns nothing the benchmark does not already deliver.

The table below applies the same statistics and ordering used for the raw returns in Section 3 to the benchmark-orthogonalized return paths above.

Portfolio	Months	Ann. ret.	CAGR	Ann. vol.	Sharpe	t
Combined model	197	1.50%	-0.00%	17.42%	0.09	0.34
External Finance	197	4.35%	2.89%	17.38%	0.25	0.96
Investment	203	1.26%	-0.34%	17.85%	0.07	0.28
Existing model	230	5.15%	5.02%	6.91%	0.75	3.24

5 Attribution to configured factor controls

A different question on different regressors: not whether the book beats the SPY market benchmark, but what the return is made of. Each raw net return is regressed simultaneously on VAL+MOM+BAB+QMJ, $r_{p,t} = \alpha + \sum_k \beta_k f_{k,t} + \varepsilon_t$. The first table gives the exposures themselves: each cell is β_k with its HAC t in brackets.

Portfolio	VAL	MOM	BAB	QMJ
Combined model	0.56 (4.66)	-0.06 (-0.63)	-0.10 (-1.16)	0.32 (3.15)
External Finance	0.42 (3.54)	-0.04 (-0.41)	-0.13 (-1.48)	0.42 (3.72)
Investment	0.52 (4.27)	-0.10 (-0.85)	-0.11 (-0.99)	0.27 (2.51)

The second table turns those exposures into return, $\beta_k \times 12 \times \overline{f_k}$. Explained sums the factor contributions; with α they account for Total, so each row accounts for the whole return. A book whose total is almost entirely one factor's contribution is that factor relabelled.

Portfolio	VAL	MOM	BAB	QMJ	Explained	Total
Combined model	-1.11%	-0.50%	-0.55%	2.25%	0.08%	-1.27%
External Finance	-0.83%	-0.30%	-0.72%	3.03%	1.18%	2.08%
Investment	-0.91%	-0.78%	-0.54%	1.95%	-0.28%	-2.39%

5.1 Simultaneous factor spanning and factor-orthogonalized returns

This is the same simultaneous factor regression expressed as a spanning test. The column headed "R2: VAL+MOM+BAB+QMJ" is the share of each portfolio's monthly net-return variance jointly explained by those factors. It is separate from the benchmark R2 in Section 4.

Portfolio	N	R2 (4F)	F
Combined model	195	0.205	9.18
External Finance	195	0.164	4.97
Investment	201	0.163	9.57

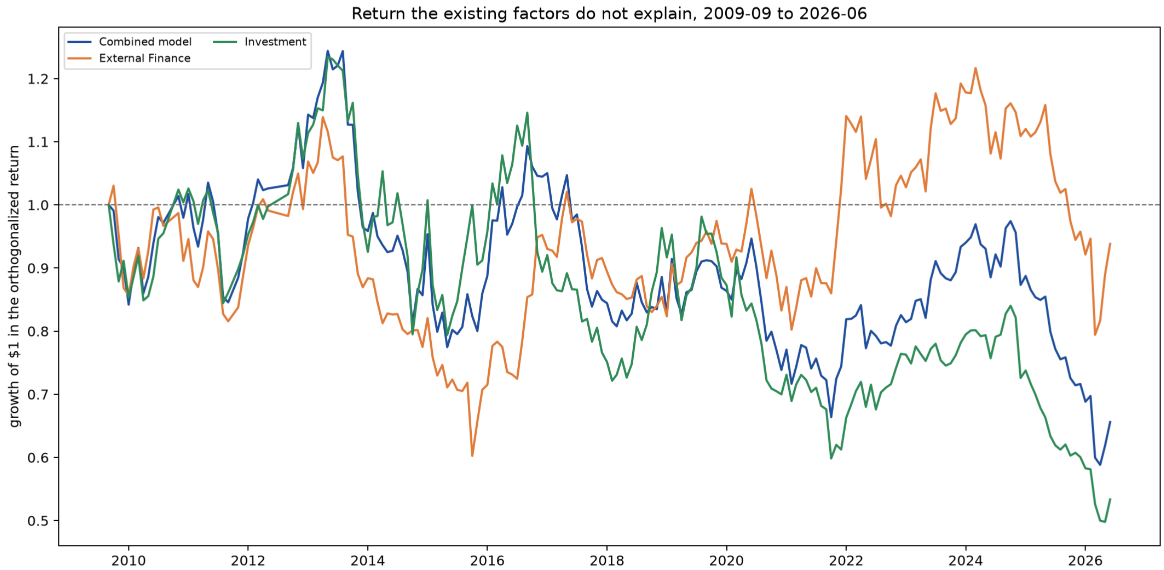

Figure 3: Returns orthogonalized to VAL+MOM+BAB+QMJ: growth of one dollar in each book once its factor exposures are removed. This is the Alpha column above, drawn as a path.

The table below applies the same return statistics used in Sections 3 and 4 to the factor-orthogonalized paths above.

Portfolio	Months	Ann. ret.	CAGR	Ann. vol.	Sharpe	t
Combined model	195	-1.36%	-2.56%	15.69%	-0.09	-0.38
External Finance	195	0.90%	-0.39%	16.04%	0.06	0.24
Investment	201	-2.11%	-3.44%	16.58%	-0.13	-0.54

5.2 The new model against the reference model

The blunt version of the same question, asked once. The combined model is regressed on Existing model taken as a single return series. The column headed "R2: Existing model" is the share of the new model's monthly net-return variance explained by that reference. A high R2 with an alpha indistinguishable from zero means the reference reproduces the new model.

Portfolio	N	Alpha	t	R2 (Existing model)	F
Combined model	195	-4.55%	-1.08	0.064	12.30

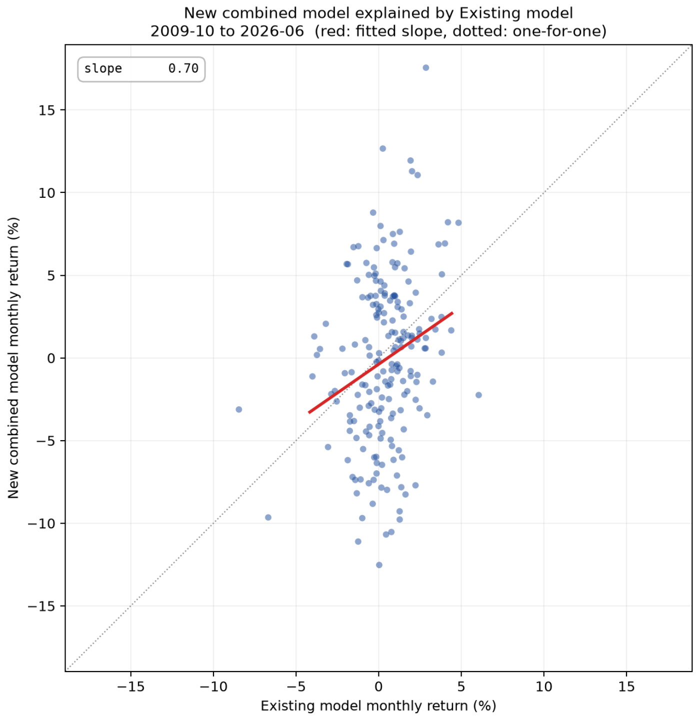

Figure 4: Each month of the combined model against Existing model, with the fitted slope. A tight cloud along the line is a model the reference reproduces; vertical spread is what it cannot.

5.3 Correlation and principal components

Whether a new sleeve is a relabelled existing factor, asked without reference to returns. The combined model is excluded: it is a blend of these same sleeves, so its correlation with its own legs is mechanical. The joint correlation and PCA span the standalone factor sleeves and the standard factor returns when supplied. The combined model is excluded: it is a blend of these same sleeve scores, so its correlation with its own legs is mechanical and carries no information about whether a sleeve is new. PC1 explains -0.3% of standardized common-window variation across 195 months. A large custom loading beside a standard factor with the same sign is descriptive overlap, not by itself proof of spanning; the regression above provides the alpha test.

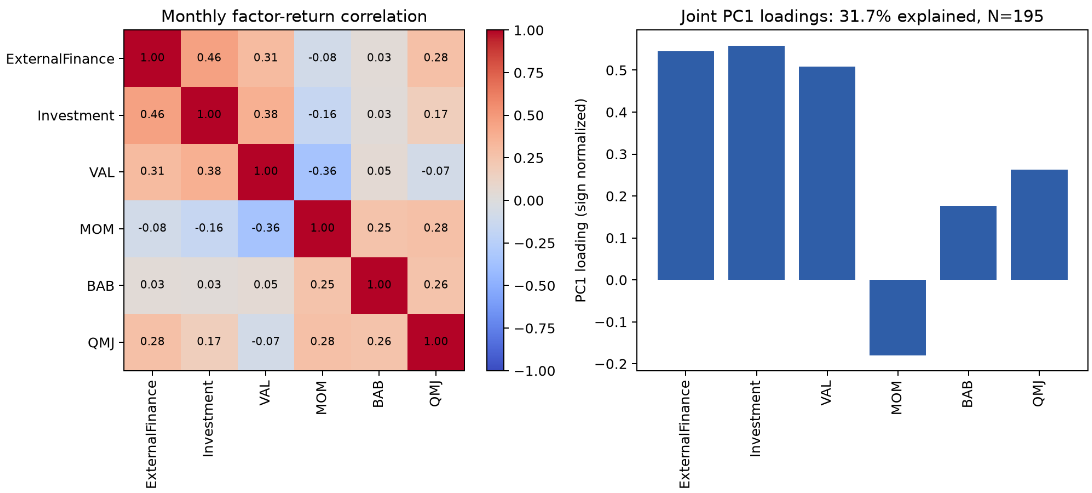

Figure 5: Monthly return correlations and sign-normalized first-principal-component loadings across the standalone sleeves and the supplied standard factors.

6 Signal persistence

How long each signal keeps predicting after formation. Every row is the mean monthly rank information coefficient between the exposure known at formation and a SINGLE later month's return – the month the portfolio trades, then one quarter, half a year and a full year out. Single months rather than cumulative windows, which would overlap and make the decay look smoother than it is. A flat profile is a slow characteristic that need not be rebalanced monthly; one that collapses after t is a fast signal whose return is being paid for in turnover.

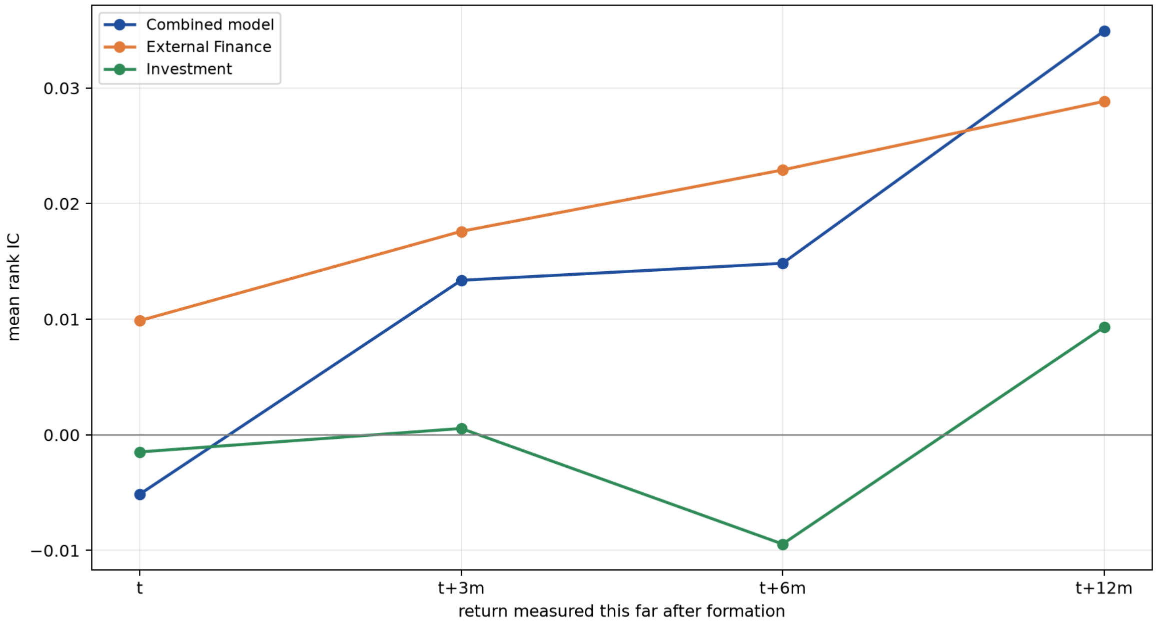

Figure 6: Mean rank information coefficient at each horizon. A line that stays flat is a signal that keeps working long after the month it is traded.

Portfolio	**Horizon**	**Mean IC**	**t**	**Months**
Combined model	t	-0.0051	-0.33	197
Combined model	t+3m	0.0134	0.74	173
Combined model	t+6m	0.0148	0.79	166
Combined model	t+12m	0.0349	2.15	158
External Finance	t	0.0099	0.62	197
External Finance	t+3m	0.0176	1.04	173
External Finance	t+6m	0.0229	1.30	166
External Finance	t+12m	0.0288	1.82	158
Investment	t	-0.0015	-0.09	219
Investment	t+3m	0.0005	0.03	210
Investment	t+6m	-0.0094	-0.60	206
Investment	t+12m	0.0093	0.57	197

7 Candidate comparisons and replacement

Each configured pair is judged twice, and the two answers can disagree. Standalone asks whether the candidate is the better factor. In model asks the question a reader with an existing model actually faces – the existing model rebuilt with the baseline removed and the candidate in its place – because a candidate that is individually worse can still improve the book if it is less correlated with what is already there. Change is against the incumbent row for a candidate, and against the unmodified existing model for an in-model row.

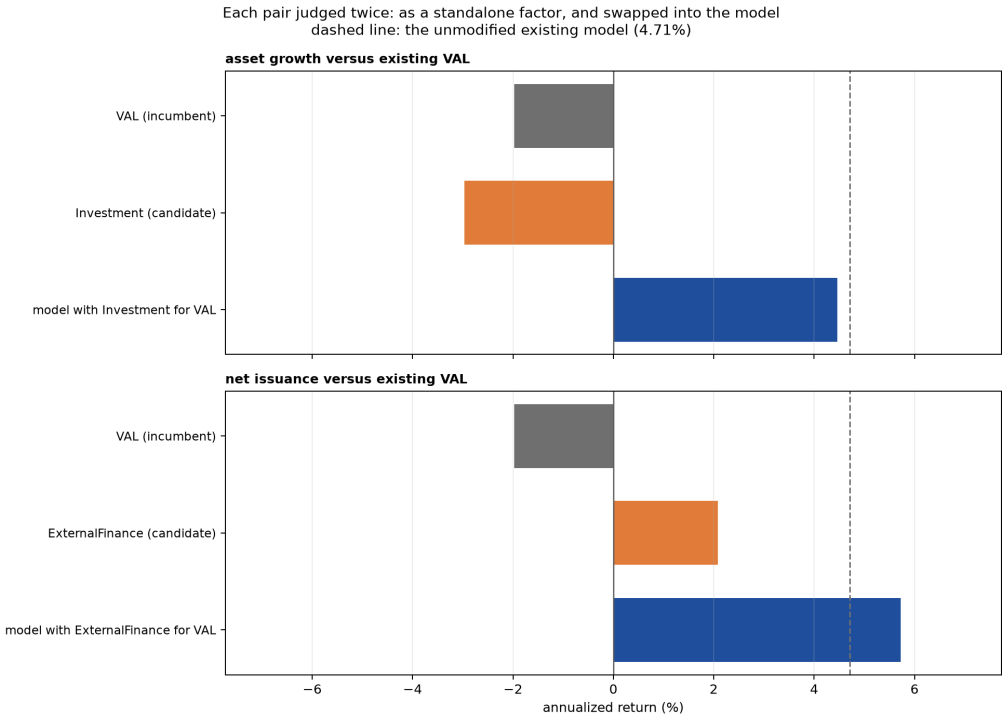

Figure 7: Each pair judged twice: the incumbent factor against the candidate standing alone, then the model with the candidate swapped in. The dashed line is the unmodified existing model.

Pair	Role	Portfolio	Ann. ret.	Change	Alpha	Change	t
asset growth versus existing VAL	incumbent	VAL	-1.97%	–	-1.49%	–	-0.36
	candidate	Investment	-2.97%	-1.00%	0.59%	+2.08%	0.12
	in model	Existing model, Investment for VAL	4.47%	-0.25%	6.33%	+0.52%	3.49
net issuance versus existing VAL	incumbent	VAL	-1.97%	–	-1.49%	–	-0.36
	candidate	External Finance	2.08%	+4.05%	4.04%	+5.53%	0.85
	in model	Existing model, External Finance for VAL	5.73%	+1.01%	7.19%	+1.38%	3.88

8 Conclusion

The run combines 2 normalized factor scores (Investment, ExternalFinance) into one stock-level model score before constructing one portfolio on 300 sampled symbols from gold version=20260822T141457Z. The combined net model earns -1.09% annually on an arithmetic basis and -2.61% compounded (the figure the growth chart shows), with 17.64% volatility, Sharpe -0.06, and HAC t=-0.24. Two questions decide whether this model is worth holding. Does it earn a positive return the SPY market benchmark does not already deliver – the alpha and its p in the spanning section. And is that return simply the configured factor controls rearranged – the loadings and contributions in the attribution section. The sleeve and replacement tables are evidence for those two answers, not separate verdicts.

9 Reproducibility

Run ID:

`20260825T164802Z_capital-discipline_525a387844`

Configuration hash: `525a387844`. Gold version: `version=20260822T141457Z`. The run directory contains normalized configuration, factor and model exposures, the combined model return, attribution-sleeve returns, statistics, regressions, figures, LaTeX source, compiler log, and this PDF. Re-running never overwrites an existing directory.